S07. Finite-size criteria for parent-Hamiltonian spectral gaps

Pure Thought research portfolio

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Source: synthesis challenge 23; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

A small quantum chain can have a clear energy separation above its lowest state while that separation shrinks away as the chain grows. This problem asks when an exact calculation on a short segment can rule out that collapse for an entire specified family of longer chains. A theorem connecting local and global behavior is the essential step.

Key terms

- **Quantum chain** — A one-dimensional sequence of interacting quantum degrees of freedom.
- **Matrix product state** — A structured representation of a many-body state using products of small matrices.
- **Parent Hamiltonian** — A Hamiltonian constructed so that a specified state belongs to its ground-state space.
- **Ground-state space** — The collection of states attaining the minimum energy.
- **Frustration-free** — Having ground states that simultaneously minimize every local interaction term.
- **Spectral gap** — The energy separation above the entire ground-state space.
- **Uniform gap** — A positive lower bound on that gap that does not vanish as chain length increases.
- **Thermodynamic limit** — The limit of arbitrarily large systems.
- **Finite-size criterion** — A theorem translating a suitable local calculation into a conclusion for larger systems.

Why this matters

The gap helps distinguish low-energy behavior and enters rigorous statements about correlations and stability under suitable conditions. Proving a uniform gap turns a finite numerical observation into a statement about a many-body family, which is often the scientifically relevant object.

What progress would mean

Progress would verify all assumptions of a chosen finite-size theorem and obtain a positive size-independent lower bound. Failing its threshold would leave the question unresolved rather than prove gaplessness. The result would apply to the specified family, not solve the general spectral-gap problem.

Revised core question

For a specified frustration-free one-dimensional parent-Hamiltonian family, can a validated local spectral calculation imply a uniform many-body gap?

Precise scope, target and outputs

Fix exact MPS tensors, on-site dimension, interaction range, boundary convention, and local positive projectors h_i. Verify frustration freeness and ground-space assumptions. Select a precise Knabe-type or martingale theorem appropriate to that model. Compute interval enclosures for the required finite-segment nonzero gaps and verify the theorem's threshold to derive a uniform lower bound for the stated sizes/boundaries. Degenerate ground spaces require identifying the first strictly positive eigenvalue rather than assuming a unique ground state.

Detailed critique of the synthesis framing

The synthesis's jump from MPS to arbitrary PEPS makes a tractable finite-size criterion look like a general spectral-gap decision method. Such general algorithms are obstructed by undecidability results. A positive finite-chain gap alone says nothing uniform as size grows; one must meet a theorem's hypotheses and threshold. The revision targets certificate automation for an explicit restricted family and preserves unresolved outcomes when the chosen sufficient criterion fails.

Validation and milestones

Baseline: certify a standard parent model with a valid finite-size theorem. Strong: a uniform parameter interval with an explicit gap lower bound. Test a deliberately gapless/threshold-failing control without claiming that criterion failure proves gaplessness.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Cubitt, Pérez-García and Wolf, Undecidability of the Spectral Gap motivates the restriction to a specified certifiable family.

Displayed formulation and claim boundary

Supplemental specification.

$$H_N = \sum_i h_i, \qquad h_i \succeq 0, \qquad \Delta_N = \min(\operatorname{spec} H_N \setminus \{0\}),$$

$$\text{target:} \quad \inf_{N \text{ in the stated family}} \Delta_N \geq \delta > 0.$$

Verify frustration freeness and ground-space assumptions. A specified finite-size gap theorem must supply the implication from validated local gaps to the displayed uniform bound. Positive gaps on finitely many chains are insufficient; failure of a sufficient criterion is not a gaplessness proof.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise “publication-quality” in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.